

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

Original Message

Message ID	<20260822094915.D95B0C004BC7@smtpgcu-27-rl.pepipost.com>
Created at:	Sat, Aug 22, 2026 at 3:19 PM (Delivered after 0 seconds)
From:	
To:	harshachowdary5927@gmail.com
Subject:	4 days left: ₹3.25L Up for Grabs A Special Pick for Our Unstop Regulars
SPF:	PASS with IP 103.52.180.164 Learn more
DKIM:	'PASS' with domain unstop.news Learn more
DMARC:	'PASS' Learn more

[Download Original](#)

[Copy to clipboard](#)

```
Delivered-To: harshachowdary5927@gmail.com
Received: by 2002:a05:7108:4a05:b0:580:7b58:1fbf with SMTP id dt5csp1965123gdb; Sat, 22 Aug 2026 02:49:16 -0700 (PDT)
X-Received: by 2002:a05:6214:320a:b0:8f3:9586:c179 with SMTP id 6a1803df08f44-90c8f9f5f4emr48581256d6.17.1787392156674; Sat, 22 Aug 2026 02:49:16 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787392156; cv=none; d=google.com; s=arc-20260327;
b=NECqDHPvLRDlM0b71kIGr0ioGlduo0d1LhpaYtnwUtoQ05HUuf6eUBzUf5Yotgf3 W3oWyn9y9hWr8LcRhRpXGktZm1lH0im4xJPvxT2KYj7xsnSPAN7hmcaYakaUnYrV3iUu
dG4Xds4pFYIDX3wWd8CqZJJCmp4KHUUEpb7MzboYUrsvtb9JJZdkD76Dq6jtM7HmPHZz ++7qnWq9o282zZ2Pq95tnwWfn/boEeLAq01EX90/ZP98b1AsGRYcb7JkIXAAguPoJjdn
974n0hdz8hnyv19cZk5nFChuJRB3AI5qGngMBEQkt9nbcc/WHR1k3h6I/0jDkFUXPK4 08EQ==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=date:message-id:list-unsubscribe-post:list-
unsubscribe:feedback:id:reply-to:to:from:subject:mime-version:dkim-signature:dkim-signature;
bh=A0UpHInPMxxkVoL4qn4ZgfeK8QjYaw2KnJLW00Qcv5A-; fh=pEez344c9ilaHqKAZ09gUIC3T6tIn6y942hJ2kTCHYw-;
b=F52BmCoadRL/UIRMqa1k/KLPJ8ZY+m1fK6T24G1+c80hKbYbn8xo2GmWYfrXF8LWZp Yjrwg8d5RBF97moBnt+eKxpPvmKzBvAQHXpKe7No13jw43w1/clGnFvrd1Ivty3cNE8p
fJy61MeR3WuVnQpBzT5iulLiSAdBNCSQ5RrK518SuBn3ktjdj001ndhkgRKYjm16W5Ghr QXzL8gyTqkJEyAudyF53tdMwVqagZtFcgPYEn7b6p16bPt1M1JYyGh7Muc4Tn003sFM6
```

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

spf:delivery.unstop.news:103.52.180.164

Hide

Solve Email Delivery Problems

v=spf1 ip4:202.162.224.0/19 exists:%{ir}._spf.netcorecloud.net ~all

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	ip4	202.162.224.0/19	Pass	Match if IP is in the given range.
+	exists	%{ir}._spf.netcorecloud.net	Pass	This mechanism is used to construct an arbitrary host name that is used for a DNS 'A' record query.
-	all		Fail	Always matches. It goes at the end of your record.

Test	Result
✓ SPF Record Published	SPF Record found
✓ SPF Record Deprecated	No deprecated records found
✓ SPF Multiple Records	Less than two records found
✓ SPF Alignment	Domain found in SPF
✓ SPF Contains characters after ALL	No items after 'ALL'.
✓ SPF Syntax Check	The record is valid
✓ SPF Included Lookups	Number of included lookups is OK
✓ SPF Recursive Loop	Nor Recursive Loops on Includes
✓ SPF Duplicate Include	No Duplicate Includes Found
✓ SPF Type PTR Check	No type PTR found
✓ SPF Void Lookups	Number of void lookups is OK
✓ SPF MX Resource Records	Number of MX Resource Records is OK
✓ SPF Record Null Value	No Null DNS Lookups found
✓ SPF Authentication	SPF Passed for IP

Your IP is: 103.239.230.184 | Contact Terms & Conditions Security API Privacy Policy: (866)-494-6852 | © Copyright 2004-2026, MXToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11481738 B2

Twitter Facebook

EmailHeaders advertisement image

DMARC:unstop.news

Hide

Solve Email Delivery Problems

v=DMARC1;p=reject;pct=100;rua=mailto:dmarcreports@unstop.news

Your DNS hosting provider is "Amazon Route 53" [Need Bulk Dns Provider Data?](#)

Reported by ns-1669.awsdns-16.co.uk on 8/23/2026 at 6:03:19 AM (UTC 0), just for you.

Transcript

spf:delivery.unstop.news:103.52.180.164

Show

Solve Email Delivery Problems

Your IP is: 103.239.230.184 | Contact Terms & Conditions Security API Privacy Policy: (866)-494-6852 | © Copyright 2004-2026, MXToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11481738 B2

Twitter Facebook

dkim:unstop.news:nc2048

Hide

DKIM Public Records

v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtrrrFOTgo5SYTgnDTfKdKVyHa6oTQxCAEjgqJfmpkGstO9yD47dEPW2iQcIUl9zcFL8a0VDjg2gmp16u+Qz0ocesbaKh36RkPVo16h83ichOw+U1V2zW

DKIM Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=unstop.news; s=nc2048; h=message-id:list-unsubscribe-post:list-unsubscribe:feedback-id:reply-to:to: from:subject:mime-version:content-type;

Tag	TagValue	Name	Description
v	DKIM1	Version	Identifies the record retrieved as a DKIM record. It must be the first tag in the record.
k	rsa (Length: 2048 bits)	Key Type	The type of the key used by tag (p).
p	MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAtrrrFOTgo5SYTgnDTfKdKVyHa6oTQxCAEjgqJfmpkGstO9yD47dEPW2iQcIUl9zcFL8a0VDjg2gmp16u+Qz0ocesbaKh36RkPVo16h83ichOw+U1V2zWJcN//r/kay56ghBwQ12G8uHV3TzhAOG5453vm8XX6mNYA)kzkaCoe2/84zRwH8KXK-jpCjC6cdXk1qc37zdf sDzSP+qdo22et1LWq+SdUkoH7VEGBq+GRmAZNyVpoWOD6hEh7H2HuOgoW8Fut2Ic+AFLw00eXN1ckF5zUAnpuj0Eptaaq2Pd1h1RBubip1GRWdxja0QL0oBnT POKI7vYH4BAuChk8BzQ1DAQAR	Public Key	The syntax and semantics of this tag value before being encoded in base64 are defined by the (k) tag.

Test	Result
✓ DKIM Record Published	DKIM Record found
✓ DKIM Syntax Check	The record is valid
✓ DKIM Public Key Check	Public key is present
✓ DKIM Signature Syntax Check	The signature is valid
✓ DKIM Signature Identifier Match	Signature domain match
✓ DKIM Signature Alignment	Signature domain in alignment.
✓ DKIM Signature Duplicate Tags	Signature tags are unique
✓ DKIM Signature Expiration	The signature is not expired

Your IP is: 103.238.230.184 | Contact Terms & Conditions Security API Privacy Phone: (866)-688-6852 | © Copyright 2004-2028, MXToolbox, Inc. All rights reserved. US Patents 10839353 B2 & 11481738 B2

7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Digital Forensics E
gmail - Yahoo Indi
4 days left: €3.25L
Original Message
email header analy
Email Header Analy
Ask Gemini

mxttoolbox.com/Public/Tools/EmailHeaders.aspx?huid=aff688fe-044a-48ec-a259-11a5091b5c1f

dkim:env.etransmail.com:fnc

Hide

DKIM Public Record:

v=DKIM1; g=*; k=rsa; p=MIGfMA8GC5qGSZb3DQEBAA4GNADCBiQKBgQ0B7+oQZBQE2zK3CxcB3Q489yRvqDQp++5x5yr8CdcAoexULmGiV7nLGCPUnkXOLsNvzUoe8MzpeH7t1Vc/Xxbst8uW3t3ueMGePE+w+KchZbSeYVGtuX

DKIM Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=env.etransmail.com; s=fnc; h=message-id:list-unsubscribe-post:list-unsubscribe:feedback-id:reply-to:to: from:subject:mime-version:content-t

Tag	TagValue	Name	Description
v	DKIM1	Version	Identifies the record retrieved as a DKIM record. It must be the first tag in the record.
g	*	Granularity of the key	
k	rsa (Length: 1024 bits)	Key Type	The type of the key used by tag (p).
p	MIGfMA8GC5qGSZb3DQEBAA4GNADCBiQKBgQ0B7+oQZBQE2zK3CxcB3Q489yRvqDQp++5x5yr8CdcAoexULmGiV7nLGCPUnkXOLsNvzUoe8MzpeH7t1Vc/Xxbst8uW3t3ueMGePE+w+KchZbSeYVGtuXZc7Z88Mu/PatIM3kFRZQhx8Vwc3PbQI9x8gM7mbZzQADeD4z+3S4a6wTDAQAR	Public Key	The syntax and semantics of this tag value before being encoded in base64 are defined by the (k) tag.
d	env.etransmail.com	SDID value	The SDID claiming responsibility for an introduction of a message into the mail stream.
	unstop.news	From Domain	The domain used in the From header field.

Test	Result
❌ DKIM Signature Alignment	Signature domain not aligned.
✅ DKIM Record Published	DKIM Record found
✅ DKIM Syntax Check	The record is valid
✅ DKIM Public Key Check	Public key is present
✅ DKIM Signature Syntax Check	The signature is valid

Your IP is: 103.238.230.194 | Contact Terms & Conditions Security API Privacy Phone: (866)-698-6652 | © Copyright 2004-2025, MXToolBox, Inc. All rights reserved. US Patents 10839303 B2 & 11461738 B2

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.